On The Gravitational Nature Of Time

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Abstract

We explore a modified set of coordinate transformations that describe time, cosmic inflation and gravitational acceleration as the same process as observed from different reference frames through an unwrapped Kaluza-Klein like geometry. I'LL HAVE AI DO THIS WHEN THE PAPER'S DONE.

Keywords: gravitation, time, relativity, inflation

1 Introduction

Albert Einstein was insistent on a static Universe during the period in which he wrote his specialized theory of relativity[1–4](SR). What is proposed here is largely a re-examination of the geometry of SR under the assumption that the $\mathbb{R}^3$ space between two arbitrary points elongates as time progresses.

This dilation is then bound to γ, giving a velocity dependent function for $\vec{g}$ and a magnitude that extends into a 4th spatial coordinate, y, similar in construction to an unwrapped Kaluza-Klein (KK) dimension[5, 6] with a known, non-zero derivative with respect to y. Through these coordinate transformations time, cosmic inflation and gravitational acceleration become the same process as observed from different reference frames in a space-time geometry where the 4th spatial KK dimension acts as a concurrent density axis.

This geometry gives an alternative model of concurrency and synchronicity as a classical notion of simultaneity is maintained by attributing the additional spatial separation to the 4th spatial dimension, $dy = v\tau\sqrt{\gamma^2 - 1}$ as shown in figure 7. While maintaining classical simultaneity resolves numerous temporal paradoxes without the need for selective reference frame preference, it produces a concurrent axis that describes the ultra-dense state of the early Universe as being separated only by some Δ along this 4th spatial axis.

1.1 The Lighthouse & The Clock-Tower

Let an event be defined classically, such that a single, instantaneous event represents a local measurement as $dt \to 0$.

Let there be two overlapping $\mathbb{R}^3$ coordinate systems, **S** and **U**. Let **S** and **U** represent an equi-temporal surface within a 4-dimensional hyperspace where $y = y^{(S)}$ and $y = y^{(U)}$ in each respective 3-dimensional surface.

This gives

$$f\left(x_i, y\right) = \partial_y \mathcal{H} \tag{1}$$

where

$$f\left(x_i, y^{(A)}\right) = \mathbf{A} \tag{2}$$

for an arbitrary **A** contained within $\mathcal{H}$.

Let **U** represent a 'future' state from the perspective of an observer in **S** so that

$$v^{(S)} = (t, x_\mu) \qquad v^{(U)} = (t + \Delta t, x_\mu) \tag{3}$$

where $\Delta t > 0$. Likewise, let **S** and **U** be bound such that

$$f\left(x_i, y^{(U)}\right) = f\left(x_i, y^{(S)}\right) + \int_{x_0^{(S)}}^{x_0^{(U)}} F\left(x\right)\ \mathrm{d}t \tag{4}$$

where F is the sum of all forces acting on the point of space at x, integrated according to time, which in turn sums the change between the temporal surfaces **S** and **U**.

Let there also be two observers, A and B. Let A and B agree that B should travel at some *agreed upon* relativistic velocity v' between two arbitrary points in space contained within $\mathcal{H}$, p_1 and p_2. Let the experiment be arranged such that B passes p_1 in **S**, and p_2 in **U**.

Let A remain at relative rest with respect to p_1, and at p_2 place a time keeping device large enough to be visible from p_1.

We then make these assertions

1. To maintain a classical model of simultaneity A and B must agree upon the time at which B reaches p_2.
2. To satisfy existing experimental evidence for the additional γ factor, $\Delta t = \gamma \frac{d}{v}$.

If A and B are to disagree upon the time at which B reaches p_2 while requiring that $v = v'$, this single, instantaneous event (B reaching p_2) must occur more than once. For this reason we theorize that A and B *must* agree upon the time at which B reaches p_2 as a fundamental property of our reality.

As the additional factor of γ has been demonstrated experimentally on a number of occasions[7–11], we should conclude that both A and B observe B reaching p_2 in $\frac{d}{v}\gamma$. This gives a set of coordinate transformations such that for each relative observer

$$\Delta t = \overbrace{\frac{d \cdot \gamma}{v^{(U)}}}^{B} = \underbrace{\frac{d}{v^{(S)}} = \frac{d}{v^{(U)}\left(\frac{1}{\gamma}\right)}}_{A} \tag{5}$$

2 Theoretical Framework

We acknowledge two primary properties of time:

1. A driving rate of change, as fundamental as the $\mathbb{R}^3$ dimensions.
2. A 4th displacement coordinate.

Throughout the following derivations, we presume that time has no physical significance beyond the two properties described above. We presume that time acts only as a mathematical tool: $\Delta t = \frac{d}{v}$. In turn, we must attribute the 4th displacement coordinate to a position along the density axis that arises, y, and the driving rate of change to *motion*.

Likewise, we presume that in a co-moving frame of reference along the y axis, mechanical equations should limit at their Newtonian counterpart with only the relativistic portion of associated magnitudes protruding out of the containing temporal surface.

To remove time as a physically meaningful quantity, we let

$$\tau = \frac{\dot{x}}{\widehat{v}} \tag{6}$$

where $\dot{x}$ indicates not a derivative with respect to time, but rather a fundamental driving rate of change. Let $\widehat{v} = v \cdot 1t$, or the velocity in units of distance.

2.1 On Lorentz Invariance

We consider these coordinate transformations with regards to the Lorentz transformations and find that for an instantaneous event where $dt \to 0$

$$\boxed{\begin{aligned} dx' &= \gamma dx = v^{(U)} \\ c\,dt' &= -\gamma \frac{v}{c} dx = -\partial_\tau y \end{aligned}} \tag{7}$$

where y is as defined in equation 42.

By scaling $\partial_x y$ by $v\,(\gamma)$ we produce a closed symmetry such that

$$\partial_x y \; v\,(\gamma) = \partial_\tau y \tag{8}$$

where

$$v\,(\gamma) = c\sqrt{1 - \frac{1}{\gamma^2}} \tag{9}$$

2.2 The Equivalence Principle

Here we examine these coordinate transformations in the context of the equivalence principle. While a well defined function remains elusive, a numerical intersection with the fine structure constant allows us to derive several important properties of an apparent electromagnetic function producing $\vec{g}$.

2.2.1 Earth's Kinematic Velocity From Electromagnetic Parameters

We define $\delta(M,R)$ to be

$$\delta(M,R) = \frac{d}{dR}g = 2\frac{GM}{R^3} \tag{10}$$

If $\delta(M,R)$ is induced by spatial dilation, we should presume that this dilation is radial. We then find the magnitude of a single $\mathbb{R}^3$ axis by taking the inverse of the determinant of a diagonal 3×3 matrix:

$$\omega(M,R) = \sqrt[3]{\frac{d}{dR}g} = (\delta(M,R))^{1/3} \tag{11}$$

Investigating this quantity for Earth gives

$$\omega_{\oplus} = 0.014537\ldots \tag{12}$$

While this approximates 2α, it more closely produces

$$\omega_{\oplus} = \alpha(2-\alpha) \pm 0.0253\% \tag{13}$$

Then by expanding equation 13

$$\alpha(2-\alpha) = \sqrt[3]{\frac{d}{dR}g} \tag{14}$$

Moving the electromagnetic parameters and R to one side before integrating with respect to R gives

$$R^4\left(\frac{e^6\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{4h^3} + \frac{3\mu_0 e^{10}\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{64\epsilon_0 h^5} - \frac{3\mu_0^2 e^8}{16\epsilon_0^2 h^4} - \frac{\mu_0^3 e^{12}}{256\epsilon_0^3 h^6}\right) = 1.272\times 10^{21} \tag{15}$$

We then take the radius independent magnitude, isolating the additional electromagnetic parameters

$$\frac{e^6\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{4h^3} + \frac{3\mu_0 e^{10}\left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{64\epsilon_0 h^5} - \frac{3\mu_0^2 e^8}{16\epsilon_0^2 h^4} - \frac{\mu_0^3 e^{12}}{256\epsilon_0^3 h^6} \tag{16}$$

As γ represents 1 plus some small ds, let us add 1 to these additional parameters and substitute γ_{em} into equation 9.

$$\gamma_{\text{em}} = 1 + \frac{e^6 \left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{4h^3} + \frac{3\mu_0 e^{10} \left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{64\epsilon_0 h^5} - \frac{3\mu_0^2 e^8}{16\epsilon_0^2 h^4} - \frac{\mu_0^3 e^{12}}{256\epsilon_0^3 h^6} \tag{17}$$

$$v_{(\gamma_{\text{em}})} = 371,721 \text{ m/s} \tag{18}$$

This matches the 2018 Plank data[12] to within 0.514%, and is in rough agreement with general observation of the velocity necessary to induce the CMB dipole.

2.2.2 Solar ω

We expand equation 14 and define $\Gamma(\sigma)$ to produce

$$\alpha = \Gamma(\sigma) \left(\frac{d}{dR} g\right)^{1/3} \tag{19}$$

Where $\Gamma(\sigma)$ is some unknown function of spin. While a well defined function remains elusive, we can take advantage of several characteristics for specific local bodies according to table 1 to derive certain characteristics of σ.

Table 1 Advantageous properties

Body	Trait	Value
Earth	The most mass dense.	5495 kg m^{-3}
Saturn	The least mass dense.	687 kg m^{-3}
Sun	The most massive.	1.988×10^{30} kg

In equation 14 we've demonstrated that for the most mass dense body in the system

$$\Gamma_{\oplus}(\sigma) \approx \Gamma(\alpha) = \frac{1}{2-\alpha} \tag{20}$$

We propose that $\Gamma_{(\sigma)}$ acts inside of the determinant. We then take the inverse of the determinant of a diagonal 3×3 matrix

$$\Gamma_{\odot}(\sigma) \propto \sqrt[3]{\Gamma(\sigma)} \tag{21}$$

We then presume that σ is associated with spin. Due to the increased surface gravitational acceleration and the decreased rate of spin for the Sun, we approximate $\sigma = 0$.

$$\Gamma_{\odot}(\sigma) \approx \frac{1}{\sqrt[3]{2}} \tag{22}$$

Or

$$\alpha = \Gamma_{\odot}(\sigma) \sqrt[3]{\frac{d}{dR} g} = \sqrt[3]{\frac{1}{2} \frac{d}{dR} g_{\odot}} \pm 0.47\% \tag{23}$$

We then find through observation that for the least mass dense body in the system, $\Gamma_{\hbar}(\sigma) \approx 1$.

This gives a set of 3 equations where the $\Gamma(\sigma)$ term cancels for the dominant body in the system.

$$\begin{aligned} g_{\oplus} &= R_{\oplus}\left(\frac{\alpha^3(2-\alpha)^3}{2}\right) && \pm 0.008\% \\ g_{\odot} &= R_{\odot}\alpha^3 && \pm 1.426\% \\ g_{\hbar} &= \frac{1}{2}R_{\hbar}\alpha^3 && \pm 1.134\% \end{aligned} \tag{24}$$

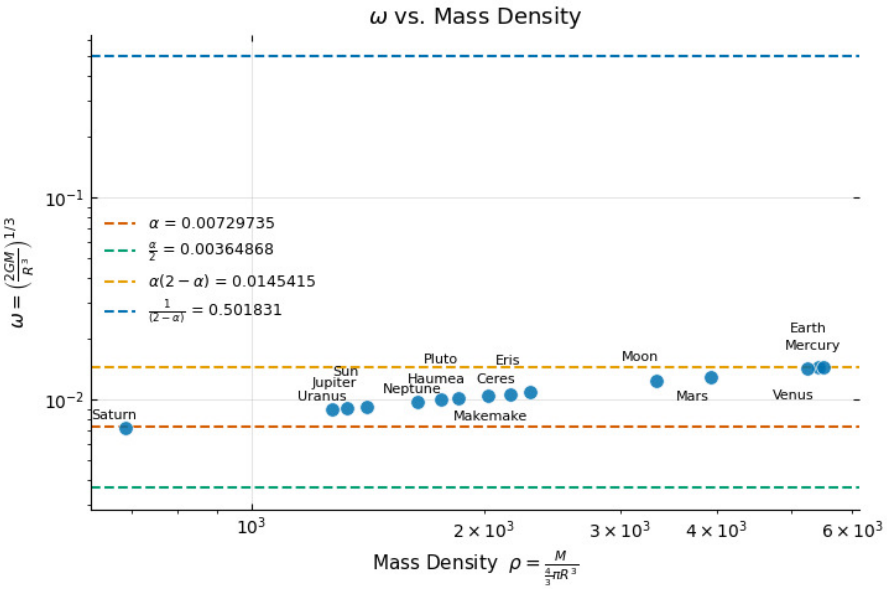


Fig. 1 The model quantity ω, identified with dy and constructed from the inverse determinant associated with volume change, plotted as a function of mass density.

Which in turn produces a domain where $\alpha \lessapprox \omega \lessapprox \alpha(2-\alpha)$ as shown in figure 1.

2.3 Gravitational Parameters

Due to algebraic symmetries, it's trivial to derive our kinematic velocity through gravitational parameters. What is of interest however, is that we derive what we propose is a more fundamental velocity first. It is only after integrating motion in a manner that is analogous to time that we produce our kinematic velocity. We call this first velocity our *cosmological velocity*, and propose a new geometric interpretation of existing data that may resolve the SNe/CMB velocity discrepancy.

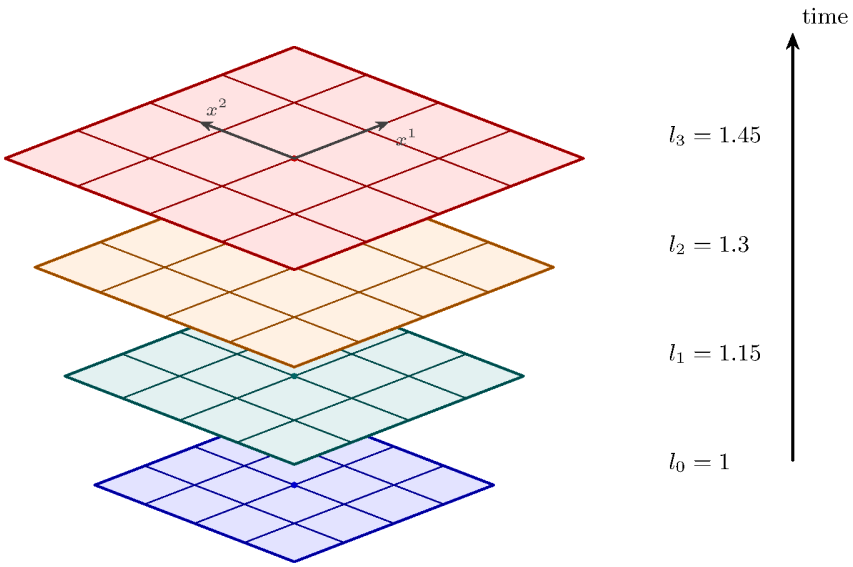


Fig. 2 Schematic stack of spatial planes with progressively larger in-plane scale factors, arranged along the time-labeled axis to illustrate the proposed concurrent density coordinate.

2.3.1 Cosmological Velocity

As γ is applied as a scalar, we take the average of the integral of $\delta_{(M,R)}$ to be

$$\varphi\left(M,R\right) = \frac{1}{R}\int \delta\left(M,R\right) = 1 + G\frac{M}{R^3} \tag{25}$$

Substituting $\varphi\left(M,R\right)$ into equation 9 gives

$$\boxed{v_{\text{cosmo}} = c\sqrt{1 - \frac{1}{\varphi^2(M,R)}} = 525,493.17 \text{ m/s}} \tag{26}$$

While this value aligns closely with the correlation carried out by Gordon et al.[13], there remains large uncertainties inherent in SNe observation[14, 15].

As shown in figure 3, this function only holds for the theoretical upper bounds of mass density. We speculate that there may exist some yet unknown process in the formation of solar systems, at least at our scale and location within the parent galaxy that leads to a natural function of mass density and velocity. While the details of this process have yet to be resolved, the coordinate transformations described here intrinsically link mass density to velocity as $\vec{g}$ becomes a function of v.

2.3.2 Kinematic Velocity

We apply the time-equivalent quantity in equation 6 to equation 25, making equation 25 time dependent. Integrating gives

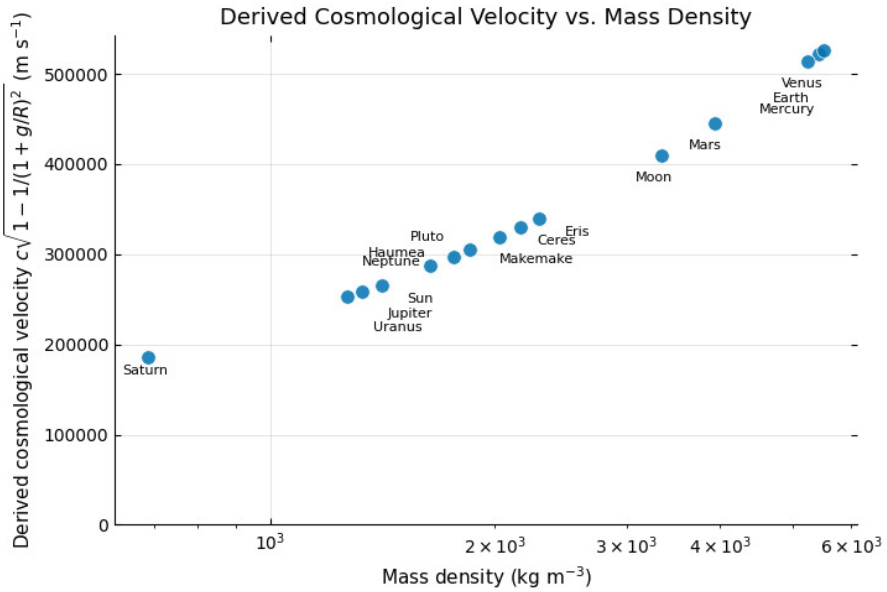


Fig. 3 Model-derived cosmological velocity plotted as a function of mass density.

$$\frac{1}{R}\int_0^R \int_0^x \delta\tau \; dR \; d\tau = 1 + \frac{1}{2}\frac{GM_\oplus}{R_\oplus^3}\left(\frac{\dot{x}}{\hat{x}}\right)^2 \tag{27}$$

We then substitute equation 27 into equation 9

$$\boxed{v_{\text{kinematic}} = c\sqrt{1 - \frac{1}{\left(1 + \frac{1}{2}G\frac{M}{R^3}\right)^2}} = 371,580 \text{ m/s}} \tag{28}$$

which matches the 2018 Plank data to within 0.47%.

2.4 Orbital Velocity

2.4.1 Relativistic Stable Orbit Velocity

In the simplified single-body system considered here, the spatial dilation described above permits a trigonometric derivation of the stable-orbit velocity as an alternative to the conventional algebraic equality.

Figure 5 represents the geometric construction used in this derivation.

$$R_0^2 + v_{\text{orbit}}^2 d\tau^2 = \left(\int_0^{R_0} 2G\frac{M}{R^3}\right)^2 = (R_0 + g \; d\tau)^2 \tag{29}$$

Setting $\tau = 1$ gives

$$\boxed{v_{(M,R),\text{orbit}} = \frac{\sqrt{GM\left(GM + 2R^3\right)}}{R^2}} \tag{30}$$

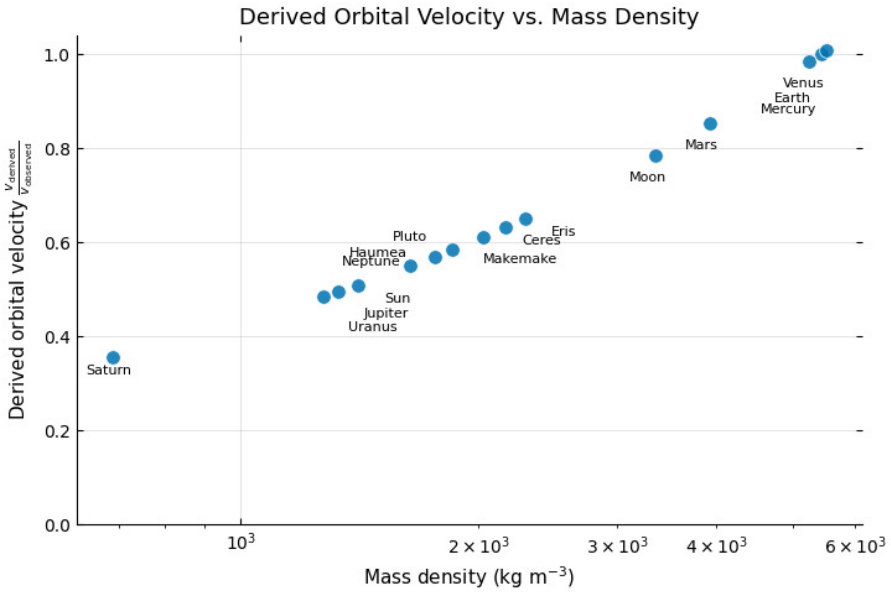


Fig. 4 Model-derived kinematic velocity plotted as a function of mass density.

For the comparison with the Milky Way, we adopt the enclosed Galactic baryonic mass reported by Cautun et al. [16]:

$$M_{\rm enc} = M_{\rm bar}(< R_0) = 4.44 \times 10^{10}\, M_\odot, \tag{31}$$

Here, $R_0 = 8.12\,\mathrm{kpc}$ denotes the Sun's Galactocentric orbital radius[17, 18].

For comparison with observation, we use the solar-neighborhood circular velocity reported by Eilers et al. [19]. Using the adopted enclosed mass and solar orbital radius, the model gives the following orbital velocity:

$$v_{(M_{\rm enc}, R_{\rm orbit}),\rm orbit} = 216,874\ \mathrm{m/s} = v_{\rm eilers} \pm 5.59\% \tag{32}$$

The model velocity equals the comparison velocity at $M_{\rm enc} \approx 4.95 \times 10^{10} M_\odot$.

The adopted mass scale is also consistent with the characteristic stellar mass scale of approximately $5 \times 10^{10}\, M_\odot$ identified by Yu et al. [20] in their FIRE-2 simulations of Milky Way-mass galaxies. Yu et al. associate this scale with a transition in galactic morphology from predominantly disc-dominated to increasingly spheroid-dominated systems. This result provides additional context for the adopted Galactic mass parameter.

2.4.2 Kinematic Velocity From Triginometric Orbit Equations

We now investigate equation 32 by setting

$$v_{\rm orbit} = \sqrt{2\frac{GM}{R}} \tag{33}$$

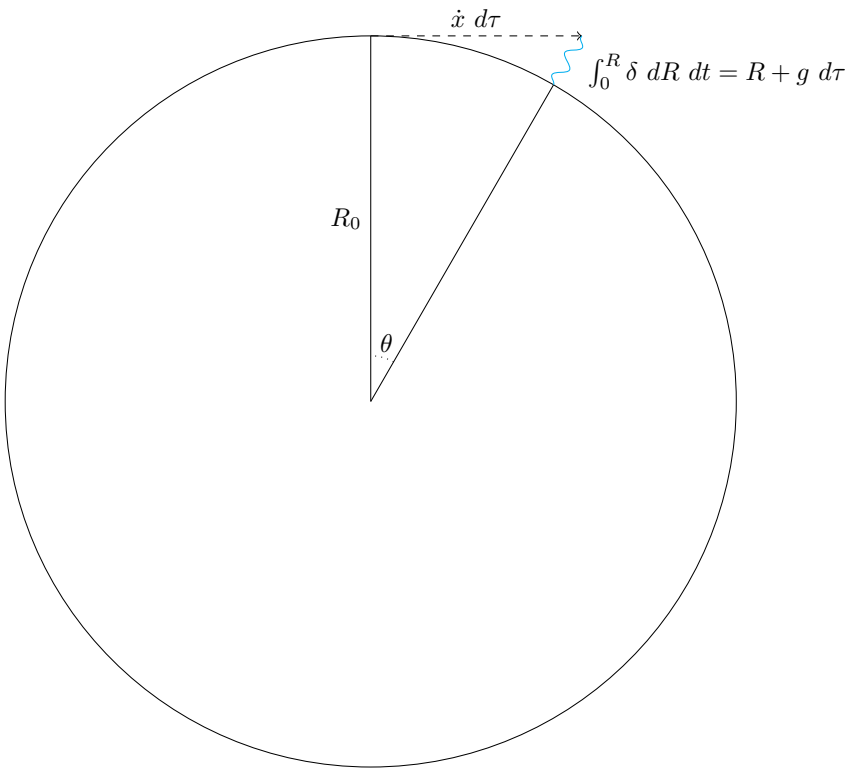


Fig. 5 Geometric construction used in the proposed stable-orbit derivation, showing R_0, the angle θ, the displacement $\dot{x}\,d\tau$, and the proposed radial increment $R + g\,d\tau$.

which is obtained by $E_{\text{kinetic}} = \int g\ dR$. Solving equation 29 for τ and substituting v_{orbit} gives

$$\mathrm{d}t = \frac{2R^3}{GM - 2R^3} = -1.000000768 \tag{34}$$

Inserting this quantity into equation 9 gives

$$v_{(|\mathrm{d}t|)} = 371,580 \text{ m/s} \tag{35}$$

which matches the 2018 Plank data to within 0.47%.

2.5 Fifth-Dimensional Spatial Displacement and Gravitational Force

While most of the derivations to this point have relied on simplified single body solutions, we explore the connections to the KK 5-dimensional metric here. As the KK model imposes a cylindrical condition largely due to the lack of an observable 4th spatial dimension, we derive the magnitude of this 4th dimension from the coordinate transformations described above and apply no such condition.

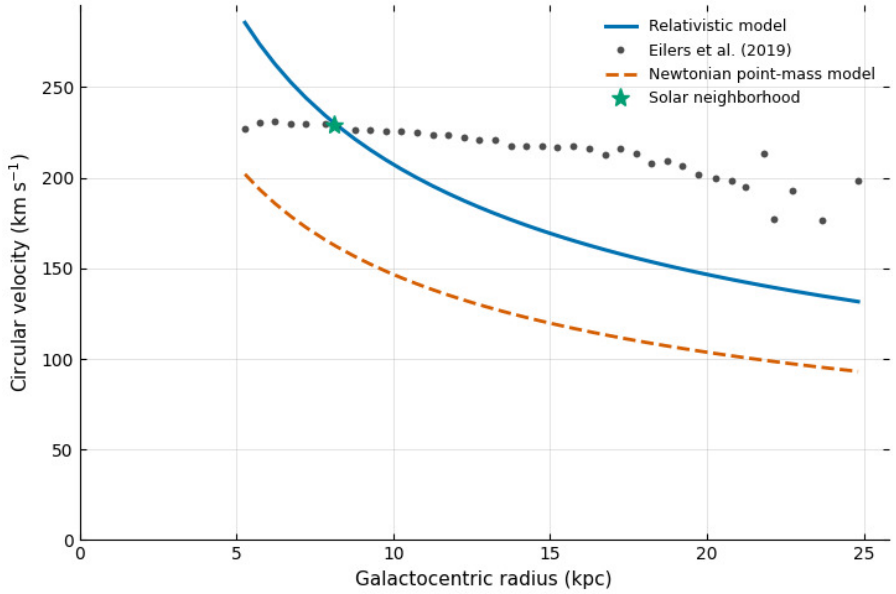


Fig. 6 Milky Way circular velocity as a function of galactocentric radius. Eilers et al. (2019) measurements are shown as points, alongside the relativistic and Newtonian point-mass model curves. The solar-neighborhood measurement is marked with a star; error bars indicate the reported uncertainties in galactocentric radius and circular velocity.

2.5.1 Geometric Construction of the Fifth-Dimensional Displacement

We extend the ordinary three-dimensional displacement into the additional spatial coordinate y. Let dx denote the magnitude of the ordinary spatial displacement,

$$dx = \sqrt{dx_i dx^i}, \qquad i = 1, 2, 3. \tag{36}$$

For spatial velocity magnitude v and displacement parameter τ, as defined in the theoretical framework, the ordinary displacement is

$$dx = v\tau. \tag{37}$$

The proposed construction treats the extended displacement $dx^{(U)}$ as the hypotenuse of a right triangle whose perpendicular components are dx and dy as shown in figure 7. With γ denoting the Lorentz factor defined in Equation 9,

$$dx^{(U)} = v\tau\gamma. \tag{38}$$

Applying the Pythagorean theorem,

$$\left(dx^{(U)}\right)^2 = dx^2 + dy^2, \tag{39}$$

substitution gives

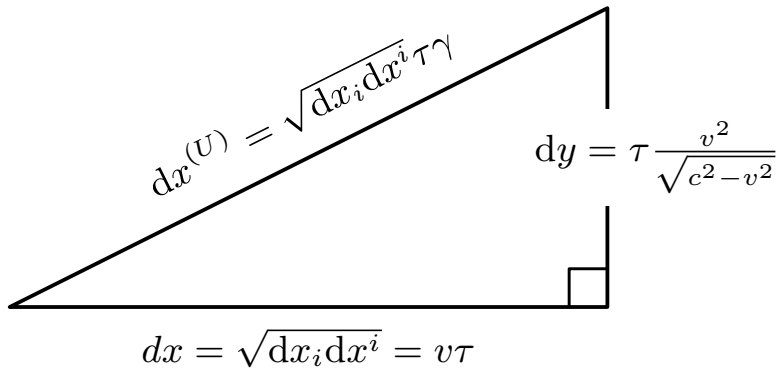


Fig. 7 Right-triangle construction used to represent the ordinary spatial displacement dx, the additional spatial displacement dy, and the extended displacement $dx^{(U)}$. The relation assigned to the hypotenuse is an assumption of the proposed geometric construction.

$$dy^2 = (v\tau\gamma)^2 - (v\tau)^2 \tag{40}$$

$$= v^2\tau^2\left(\gamma^2 - 1\right). \tag{41}$$

Taking the positive magnitude of the additional displacement yields

$$dy = v\tau\sqrt{\gamma^2 - 1} = \tau\frac{v^2}{\sqrt{c^2 - v^2}}. \tag{42}$$

The velocity form follows from the Lorentz-factor relation in Equation 9.

For constant v, differentiation with respect to τ gives the rate of displacement along y:

$$\boxed{\partial_\tau y = \frac{v^2}{\sqrt{c^2 - v^2}}} \tag{43}$$

2.5.2 Incorporation into the Kaluza–Klein Metric

We represent the additional spatial coordinate through the metric component G_{44}. For a diagonal five-dimensional metric, the line element is

$$dS^2 = g_{\mu\nu}dx^\mu dx^\nu + G_{44}dy^2. \tag{44}$$

The corresponding metric tensor has the block structure

$$\boxed{G_{AB} = \begin{pmatrix} g_{\mu\nu} & 0 \\ 0 & G_{44} \end{pmatrix}} \tag{45}$$

The component G_{44} sets the metric scaling of the additional coordinate. Its radial variation determines the coupling between motion along y and radial motion in the ordinary spatial coordinates.

For radial coordinate R, the Christoffel symbol associated with this coupling is

$$\Gamma^R_{44} = -\frac{1}{2} g^{RR} \frac{\partial G_{44}}{\partial R}. \tag{46}$$

For a normalized radial coordinate, $g^{RR} = 1$, so

$$\boxed{\Gamma^R_{44} = -\frac{1}{2} \frac{\partial G_{44}}{\partial R}} \tag{47}$$

2.5.3 Finding G_{44}

We take

$$\partial_x y = \frac{v}{\sqrt{c^2 v^2}} \tag{48}$$

Or

$$dy = \frac{v}{c^2 - v^2} dx \tag{49}$$

We then take the KK metric to be

$$dS^2 = g_{\mu\nu} dx^\mu dx^\nu + G_{44} dy^2 \tag{50}$$

and the $\mathbb{R}^4$ spatial separation to be

$$dS^2_{\text{spatial}} = dx^2 + dy^2 \tag{51}$$

where $dx^2 = dx_i dx^i$ for $i = 1, 2, 3$. Expanding this gives

$$dS^2_{\text{spatial}} = dx^2 \left(1 + \frac{v^2}{c^2 - v^2}\right) = dx^2 \gamma^2 \tag{52}$$

which matches the expected displacement magnitude of the Lorentz transformations.

Applying a flat metric to the line element defined in equation 50 gives

$$dS^2_U = dx^2 + G_{44} dy^2 \tag{53}$$

yielding

$$\boxed{G^{(U)}_{44} = 1} \tag{54}$$

for $dS^2_U = \gamma^2 dx^2$.

However solving equation 53 for G_{44} in the **S** coordinate system where $dS^2_S = d_i d^i$ gives

$$\boxed{G^{(S)}_{44} = 0} \tag{55}$$

3 Discussion

3.1 The Appearance of Electromagnetic Parameters

We do not claim to derive a direct mechanical derivation for the fine structure constant, nor an explanation for it's appearance in section 2.2. We simply apply the coordinate transformations defined in section 2 and discover that α appears repeatedly. While the probability is difficult to quantize, we feel that this repeated appearance of α, appearing in gravitational approximations for all 3 of the local bodies with boundary like characteristics likely indicates a shared function which itself contains or produces α. Due to the hypersphere-like geometry that arises from this model, we propose further investigation into the boundary between Wyler's derivation[21] of α and the $4+0$ space-time described here.

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